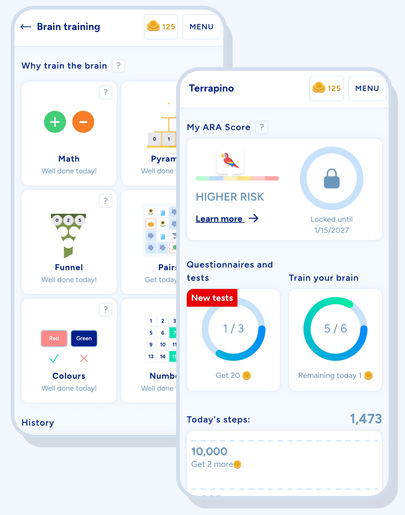
Terrapino

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# Digital Cognitive Training and Dementia Risk Monitoring: Real-World Evidence from the Terrapino Mobile App





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**A gamified brain-health app reaches the people dementia prevention needs to reach: two in three of 38,411 real-world users are aged 50–80.**

**38,411** users, 3 years of real-world app data (Czech & English)

**67%** aged 50–80, the core target group for prevention

**36%** of 18,733 risk-assessed users in an increased-or-higher risk category

**41%** below-normal memory score on the Harrison BASIC test

## 1. Background

- Prevention is a major factor in reducing disease onset; modifying 14 risk factors could prevent or delay up to 45% of dementia cases worldwide.[1]
- In addition to activation through specific cognitive exercises, physical activity (such as walking) is crucially important, as maintaining physical activity protects against dementia.
- Prevention needs tools that reach people **before symptoms**, at population scale: risk screening, cognitive testing and lifestyle support.
- Mobile apps can do this, but **keeping users engaged** long term is the main barrier in digital health.

## 2. Objective

- To evaluate user uptake and long-term engagement with the Terrapino mobile app.
- To characterize the data streams generated by the platform.
- To demonstrate how the app supports dementia risk screening, cognitive training, and healthy lifestyle habits.

## 3. Methods

- **Design:** retrospective analysis of anonymized real-world data from **38,411** Czech- and English-speaking users over 3 years.
- **What the app measures and trains:**
  - **ARA score:** Alzheimer's Risk Assessment questionnaire on modifiable risk factors, physical activity and diet.
  - **Harrison BASIC:** repeatable episodic-memory test (Brief And Simple Index of Cognition, with Prof. John Harrison).
  - **Emotion recognition** task (social cognition).
  - **Daily training:** 7 cognitive mini-games and a step tracker for walking.
- **Motivation:** points for activity, redeemable for partner offers.
- **Outcomes:** uptake, engagement, risk categories, cognitive test results.

## 4. Results

**Fig 1. Two in three users are aged 50–80**

The app reaches the age group at increased risk of cognitive decline, not only younger, tech-savvy users.

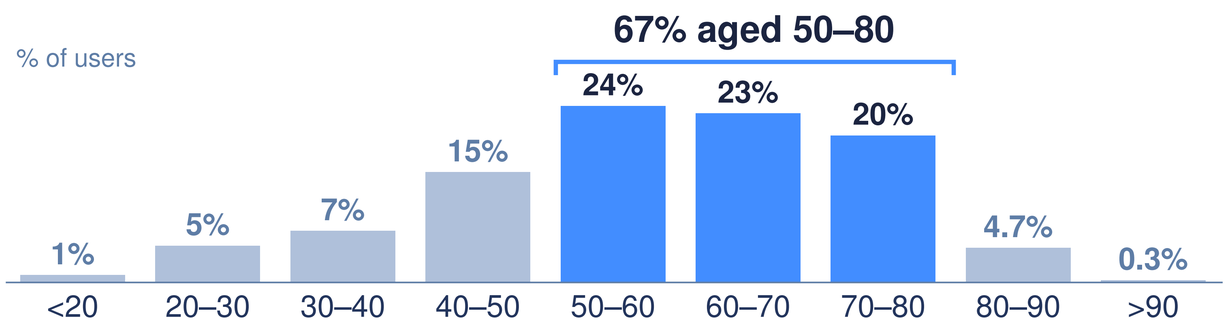
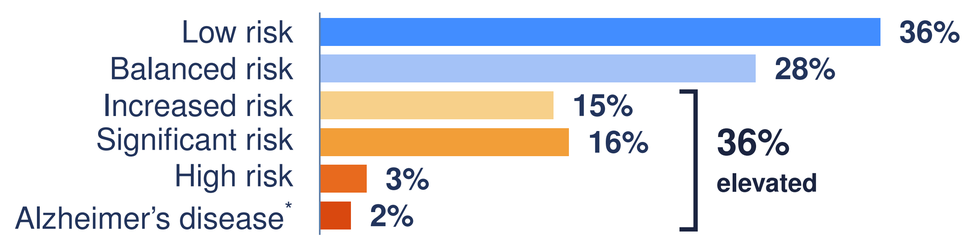



**Fig 2. One in three risk-assessed users has elevated risk**

**48.8%** of users (n = 18,733) completed the ARA questionnaire at least once.




*Name of the highest ARA questionnaire category; not a clinical diagnosis.

**Fig 3. Users and engagement**

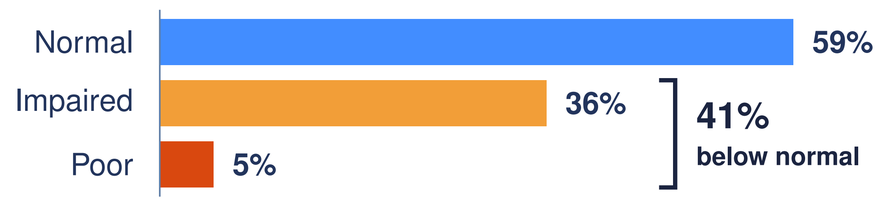



**10:33 min** mean session length, highly active users

**1,038** daily active users of the 7 games (last 6 months)

**Fig 4. 41% scored below normal on a memory test**

Harrison BASIC, ≈4,500 users tested. A screening result in a self-selected sample, to be confirmed clinically.




## 5. Conclusion

- **Target population reach:** a gamified app reached 38,411 users, two thirds of them in the 50–80 age group that prevention targets.
- **Risk is visible early:** one in three risk-assessed users had an increased-or-higher risk profile, and 41% of tested users scored below normal on memory.
- **Users engage:** highly active users spend over 10 minutes per session; points-based rewards are a practical engagement lever.

## Next steps

- Longitudinal analysis of repeated Harrison BASIC scores and long-term retention.
- Predictive models and personalized lifestyle recommendations.
- Physician reports that bring app data into clinical care.
- More language versions.

**App download**

**Access poster PDF**

**Medical device:** Terrapino (software), Risk class I, according to the Annex VIII, Rule 11 Regulation (EU) 2017/745 of the European Parliament and of the Council on medical devices.


**Reference:** 1. Livingston G, Huntley J, Liu K et al. Dementia prevention, intervention, and care: 2024 report of the Lancet standing Commission. The Lancet, 2024; 404, 572–628.